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Introduction

Mexico City is the economic powerhouse for the country, and is the core of the federal government. Scattered throughout the city there are not only headquarters and offices for all the federal executive, legislative and judicial branches of the government; although embassies, organizations, media and global companies are concentrated in one single place, the daily human flow coming from other Mexican states, Central American immigrants, tourists are reflected on the streets¹, but the demographic increase occurs when that coming human flow considers settling down in the city. In 2020 the population was 9,209, 922 inhabitants², the share area has gradually increased since the mid-20th century and now subdivided into 16 delegations or administrative areas: Álvaro Obregón, Azcapotzalco, Benito Juárez, Coyoacán, Cuajimalpa de Morelos, Cuauhtémoc, Gustavo A. Madero, Iztacalco, Iztapalapa, Magdalena Contreras, Miguel Hidalgo, Milpa Alta, Tláhuac, Tlalpan, Venustiano Carranza and Xochimilco³.

Even though the downtown neighborhoods were always preferred by foreigners a long time ago, this phenomenon intensified during the lockdown due to softer COVID-19 sanitary restrictions in Mexico City than other countries, attracting digital nomads who choose the metropolis as a destination because their money had more value than in their places of origin. The expansion of mostly American or European citizens⁴ with remote work began to be noticeable specifically in neighborhoods such as La Condesa and Roma, which had a rise up in the rent and real estate pricing.

This is not new. Even that gentrification arose for the first time in certain neighborhoods of London in the 1950s, being one of the most common widespread urban phenomena throughout the world, now is impacting housing accessibility and displacing its residents in Mexico City. To demonstrate this claim through this research, we conducted a temporary analysis of house patterns in Mexico City by utilizing one single extensive database, this single data system is conformed by 16 separated databases, each one for each delegation.

This database was recovered by the Secretary of Administration and Finance (SAF)⁵. In this data set you will find the relevant information from the Mexico cadastre with the last update on September 30th, 2022. In particular, for each delegation it contains information regarding: Land area, construction surface, construction use, year of construction, unit value of land, land value, latitude, longitude, government subsidies, price key used in tax collection. The

¹ According with INEGI “movimientos migratorios”
https://cuentame.inegi.org.mx/monografias/informacion/df/poblacion/m_migratorios.aspx?tema=me
 (2020) Accessed May 27th 2024

² According with INEGI “Número de habitantes en ciudad de México”
<https://cuentame.inegi.org.mx/monografias/informacion/df/poblacion/> (2020) Accessed May 27th 2024

³ According with BRITANNICA “Mexico City”
<https://www.britannica.com/place/Mexico-City/Administration-and-society> (2021) Accessed May 27th 2024

⁴ According with Yahoo noticias “Gentrification and the real problem in CDMX that goes beyond expensive rents in Roma and Condesa”
<https://es-us.noticias.yahoo.com/la-gentrificacion-y-el-verdadero-problema-en-cdmx-que-va-mas-alla-de-rentas-caras-en-roma-y-condesa-003258838.html> (2024) Accessed May 27th 2024

⁵ SAF: a governmental institution in charge of conducting policies regarding the development of income policies and tax administration policies, programming, budgeting and evaluation of public spending in Mexico City.

cadastre study is updated every year due to tax collection reasons every January, whereby we can say that our analysis is made up from a reliable and most up to date database possible.

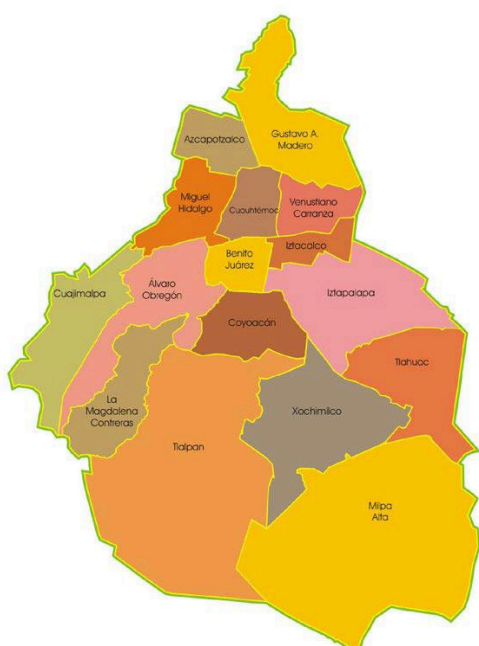
Previously to go further the model linear regression is important to get a glimpse of the Mexican City services background through the last few years. Is very well known that Mexico city provides a full range of utilities and other municipal services to wealthier and middle-class residents. However, many poorer neighborhoods lack safe drinking water, proper housing, electricity and sewer systems; Running water, electricity and paved, lit streets may also be delayed for years in some areas. Electricity is produced within the city, but most is purchased from outside.

The Stata/MP 18.0 version is used to run multiple linear regression tools to work on the data set mentioned before, which is the most recent version (est. 2023). Allowing to make an efficient use of the new graphics system and dialog boxes for all commands, those new modifications are being reflected in the next histograms with log and the different levels to check the distribution's shape of our data and their skewness.

I. Related work,

Being a new topic in the last three years, you can find a large variety of news related this topic concerning the water distribution preferences based on “fancy neighbors” but there is not a wide variety of academic research related to housing prices in Mexico city, though it may seem there are several websites to check the real estate market in Mexico city and how those differ between neighbors. Aguilar Velazquez, Rivera Islas & Romero Tecua research project is one of the most complete and similar work handled these last years. This work was based on two datasets, scraped data of 16,000 prices for flats and houses to observe temporal form 2000 to 2022 and spatial trends in Mexico City. The authors found that housing prices have quadrupled over a 20 year period, without considering inflation (in real terms). Aggravating the Mexicans' possibility to afford a house due to the per capita labor income of Mexican citizens has declined relative to inflation, they encountered four times greater challenges in accessing housing in 2015 as compared to 2005.

Cleaning data



Rather than the previous related work, in this analysis, it will be focused on gentrification impact on Mexican city house pricing, the goal here is to evidence the drastic difference between the neighbors or delegations inhabited by digital nomads or just foreigners settled in Mexico City compared with the outskirts delegations inhabited by local mexicans. To achieve this analysis purpose, it was preferred to use an official government dataset since it contains the whole population, giving more credibility to our analysis. The initial data sets were sixing separate .xlsx files containing around 15 variables in all datasets and the observations varied accordingly the delegation, for instance cuauhtémoc delegation with

252,104 observations in contrast with 8,678 observations from milpa alta delegation (which located in the border of the city, and one of the most undeveloped area).

From the beginning you can see how housing density can vary between delegations, even though that milpa alta delegation has a larger area, cuathermoc delegation has a higher housing density (see figure 1).

Because the data was scraped from an official website, there were not many cleanliness issues, but there were some variables that wasn't necessary to keep, such as [instalaciones_especiales](#) (or special facilities), [clave_valor_unitario_suelo](#) (or ground unit price key), [clave_rango_nivel](#) (or key range level) and [codigo_postal](#) (or zip code) were dropped. Leaving only [superficie_terreno](#) (or ground surface), [superficie_construccion](#) (surface built), [uso_construccion](#) (built type) , [anio_construccion](#) (year built), [valor_unitario_suelo](#) (unit ground price), [valor_suelo](#) (ground price), [subsidio](#) (subsidy), [latitud](#) (latitude) , [longitud](#) (longitude) and [colonia](#) (neighbor).

They were dropped based on the lack of influence in our dependent variable (unit ground price or [valor_unitario_suelo](#)) such as zip code, key range level which is a string variable and ground unit value key which is a key to define taxes cost; The rest of the variables were tabulated to check their influence on the ground unit value, in the specific case of [instalaciones_especiales](#) (or special facilities) was dropped, since 71.83% of cumulative percentage was "No included" in the Alvaro Obregon dataset, there was no point in keeping it, if what is going to be measured are ground unit value of residential houses not special facilities (see figure 2).

instalaciones_especiales	Freq.	Percent	Cum.
No	124,277	71.83	71.83
Si	48,745	28.17	100.00
Total	173,022	100.00	

Fig. 1

Later, it was necessary to make sure to keep only the residential homes in analysis, to be sure, the variable [uso_construccion](#) (or built type) was tabulated, the dataset was only supposed to contain housing information about homes and their respective monetary unit values, however since the urban mexico city cadastre is an inventory of real estate of urban population settlements belonging to moral and physical person, it was necessary to check the frequency of other real states that do not have of built type as housing purpose. Since there was 3.71% of "NA" observations in the variable [uso_construccion](#) (or built type) I decide to convert "NA" in empty value to later drop them, it was necessary to keep the parameters: housing(habitacional) , housing and commercial (habitación y comercial), neighborhood center (centro de barrio) , equipment (equipamiento) , green areas (áreas verdes), urban subcenter (subcentro urbano).

In Mexico it is common to have a business in the same place of housing, the majority of housing is like that, that is why it has a large frequency in our tabulation table (see figure 3.) i

keep the rest parameters since it is interesting later to compare the frequency of green areas, equipment such as hospitals, in delegations gentrified and delegations not gentrified.

```
. tab uso_construccion
```

uso_construccion	Freq.	Percent	Cum.
Centro de Barrio	686	0.40	0.40
Comercial	1,192	0.69	1.09
Equipamiento	1,638	0.95	2.03
Habitacional	81,085	46.86	48.90
Habitacional y comercial	75,532	43.65	92.55
Industrial	2	0.00	92.55
NA	6,423	3.71	96.26
Polígono de Actuación	156	0.09	96.35
Servicios Públicos	4	0.00	96.36
Sin Zonificación	2,668	1.54	97.90
Usos Existentes	60	0.03	97.93
Áreas Verdes	3,576	2.07	100.00
Total	173,022	100.00	

Fig.2

Lastly I end up with my first dataset cleaning, removing as much as possible the “NA” or any possible outlier that could be shown further in our analysis. I made sure to replace with a missing value all the “NA” found in the variable [valor_suelo](#) (ground price), which is our dependent variable and it was generated as the string length of [anio_construccion](#) (year built) and any observations with a [anio_construccion](#) (year built) greater than 4 were dropped.

In the sampling part, there were 400 observations taken for each delegation since we want to know the price housing differences between gentrified delegation and not gentrified delegations rather than know which delegations have more houses. We created a variable ID in a way to enlist the parameters for each delegation. The cleaning procedure began with Alvaro Obregon dataset, with the idea to generalize the same procedure to the remaining 15 datasets.

After having cleaned all the dataset, instead to merge the data, it was more convenient to append the 16 dataset since all of them contain the same variables and each one has 400 observations, stacking datasets will give a better handle of information, so i made use of [alvaro-obregon-cleaned.dta](#) as a master dataset and it was added some quotes and “force” command to keep the string variables.

Simple linear regression (Univariable analysis)

After the data had been sufficiently cleaned, it was ready to be used in a simple linear regression. The goal of this analysis was to observe the distributions of both the dependent and independent variables and to qualify their relationships in order to lay the groundwork for the econometric analysis and modeling; As it was mentioned previously the project is focus on make contrast the price value among delegations, is important to check how much the square meter varies along Mexico city, even there are other factors that mexicans take into

account, the squared price in real terms⁶ is the main factor when they decide to purchase a real estate property.

Hence, in the univariate analysis our dependent variable is the price which will show how the price varies and specifies where the gentrification areas are. And our dependent variable, the explanatory variable is the distances from center since many of the governmental institutions and companies headquarters are located, but first it was necessary to create a new variable which contains the distances between households from our random sample to Mexico city zocalo center, making use of latitude and longitude variables provided by dataset.

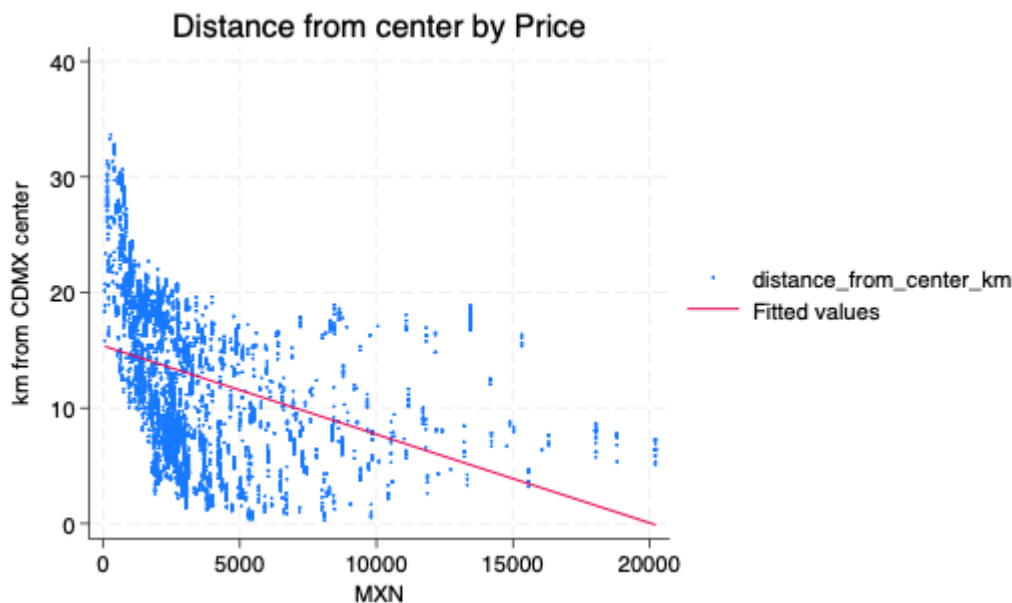


Fig. 3

The scatter plot shown above represents the relationship between the distance from Mexico city zocalo center in kilometers in relation with the ground price per meter square. The correlation among two variables is negative, if X variable increase in this case is km from center then Y variable decrease or price per meter square. The top left is empty and bottom right is where the majority of the dots are located. It can be observed that the furthest outliers in the bottom right are located less than 10 km from Mexico city center and have a price of 20,000 MXN (1,000 EUR) per meter square. And the furthest outliers on the top left of our sample has a maximum of 33.6 km from Mexico city center and has a ground price below the mean price per meter square which is 3,446 MXN (180 EUR), is important to mention that this “U” shape non linearity will be fixed later.

The distance was generated using the CDMX zocalo center latitude and longitude coordinates (-19.4326018, + 99.1332049). To convert the result of the euclidean distance formula (which are given in degrees) into kilometers, I needed to take into account the

⁶ Real term: it refers to base year prices. That is a specific year is set as a reference and the property prices of the year to be studied are taken. In this way, the properties meters squared prices are excluded from the inflation effect.

earth's curvature, so a new variable with the degrees converted into kilometers was generated, making use of the haversine formula (multiple degrees per 11.319).

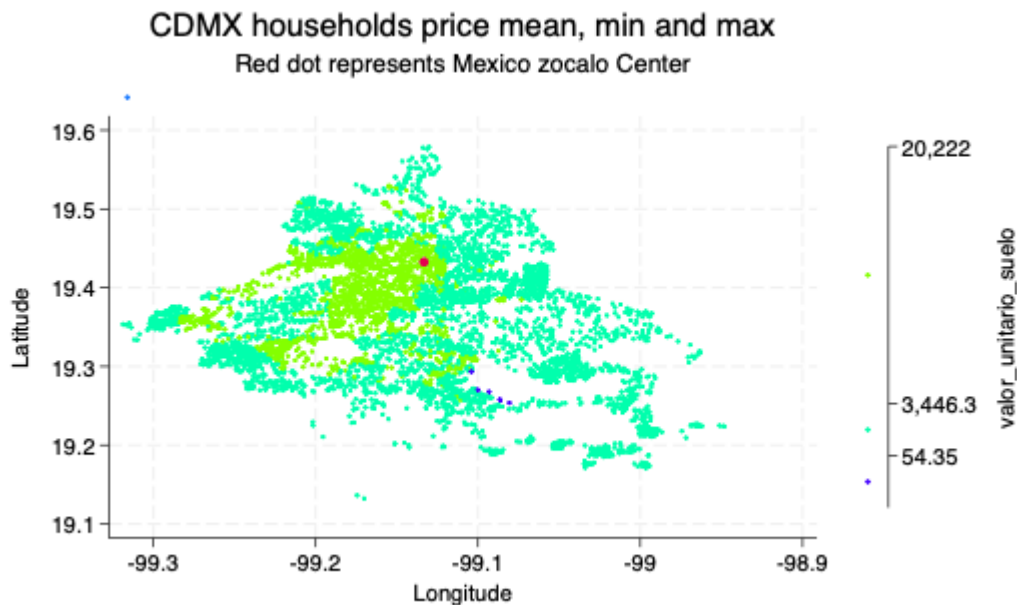


Fig. 4

To get a visual idea, where exactly households with prices above the mean are located, a twoway scatter map was generated with latitude and longitude and ground price variable ([valor_unitario_suelo](#)) display where the mean price (3,446.33 MXN), the maximum (20,222.04 MXN) and the minimum (54.35 MXN) are located, the five blue dots are located in Xochimilco. In the map above, it can be observed that the households located in the south east are well above the average.

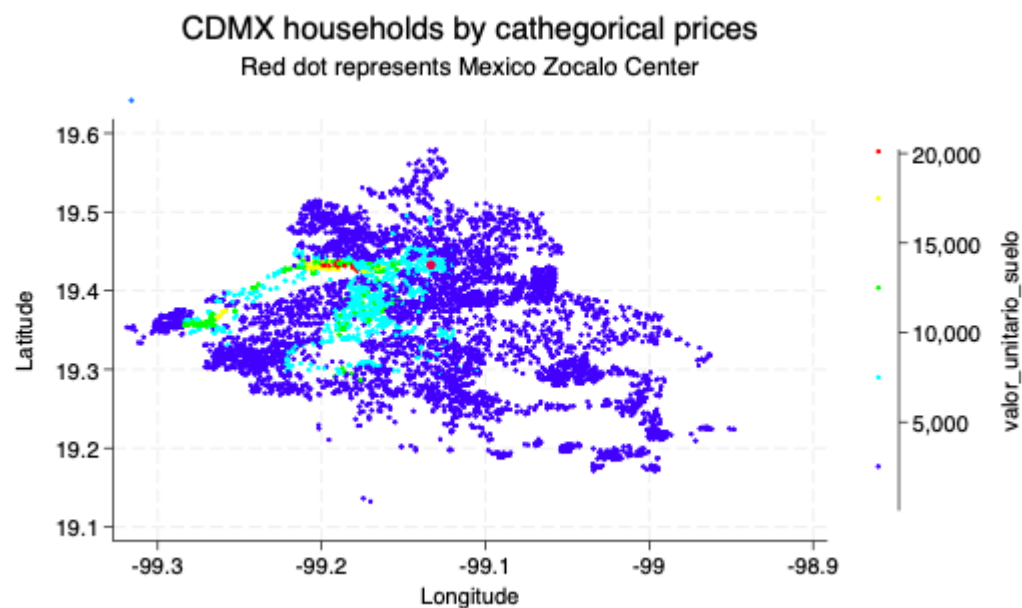


Fig. 5

In the map above, it can be observed more closely that the red small dots which represent the exacerbated prices per meter square are located in Alcandia Benito Juarez, which contains neighborhoods such as “La condesa”, “Roma”, etc. Meanwhile the turquoise blue displays the Alcaldias with greater capital gain prospects in the next years such as: Miguel Hidalgo, Cuauhtemoc, Coyoacan, Iztapalapa, Iztacalco, since they are located near to zocalo Mexico city center and where the biggest tourist attractions are located. Although many of these households will have a price restriction on real estate sales of 1 million 400 thousand MXN (70,000 EUR)⁷.

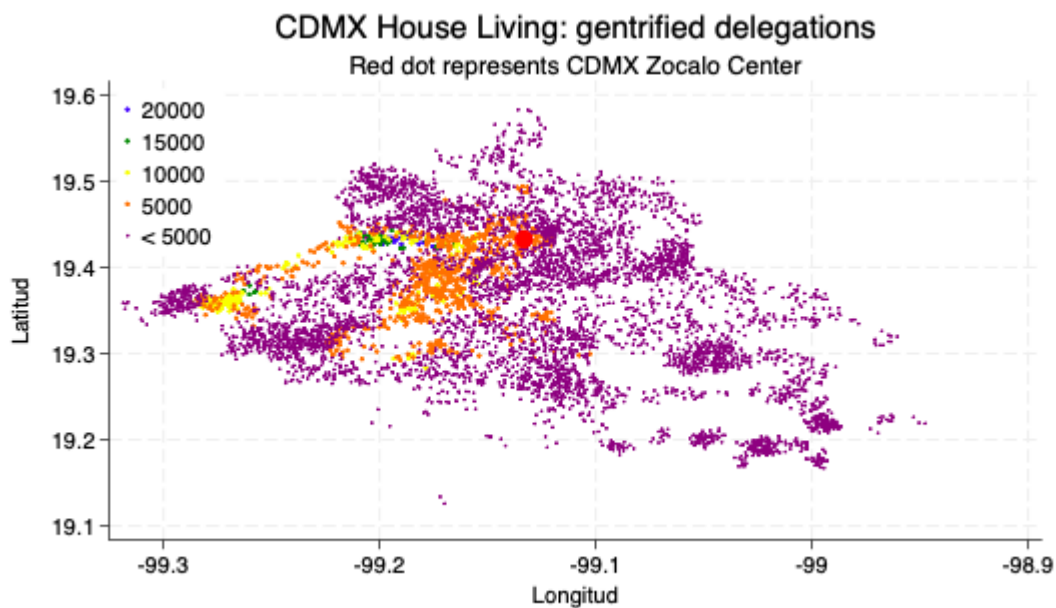


Fig. 6

Previously to go further in details and how externalities influence each delegation or Alcaldia, we had to run the first simple linear regression, where is observed that our independent variable distance from center is able to explain 11.1% of the variation of our dependent variable ([valor_unitario_suelo](#)) base on our r-squared, our p-value shows a strong significance (below 0.05 to reject H0 in both the constant and single variable, so we can say the model is significant globally.

Source	SS	df	MS	Number of obs	=	6,400
Model	7.4737e+09	1	7.4737e+09	F(1, 6398)	=	814.68
Residual	5.8693e+10	6,398	9173690.74	Prob > F	=	0.0000
				R-squared	=	0.1130
				Adj R-squared	=	0.1128
Total	6.6167e+10	6,399	10340197.6	Root MSE	=	3028.8

valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
distance_from_center_km	-147.4473	5.165859	-28.54	0.000	-157.5742	-137.3205
_cons	5324.583	75.91906	70.13	0.000	5175.756	5473.41

Fig. 7

⁷ According with Sol de Mexico “Brugada will inherit Sheinbaum’s low-cost housing projects” https://cuentame.inegi.org.mx/monografias/informacion/df/poblacion/m_migratorios.aspx?tema=me (2024) Accessed June 10th 2024

To go deeper into every delegation or Alcaldia and be able to categorize them respectively as gentrified or not, it was necessary to run again the regression model, but this time without taking into consideration the distance from center, to achieve this categorical dummies were created from 1 to 16, and we can interpret in the second regression than those Alcaldias or delegations with negative coefficient have a price opposite correlation. Whereas the following Alcaldías: Alvaro Obregon, Benito Juárez, Coyoacán, Cuajimalpa de Morelos, Cuauhtémoc and Miguel Hidalgo shows a positive relation in terms of price, interpreted as areas whose exclusivity is based on the price of the land per square meter, and therefore with capital gains for the real estate market.

```
. regress valor_unitario_suelo i.alcaldia_coding
```

Source	SS	df	MS	Number of obs	=	6,400
Model	2.8729e+10	15	1.9153e+09	F(15, 6384)	=	326.60
Residual	3.7438e+10	6,384	5864335.8	Prob > F	=	0.0000
Total	6.6167e+10	6,399	10340197.6	R-squared	=	0.4342
				Adj R-squared	=	0.4329
				Root MSE	=	2421.6

valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
alcaldia_coding						
AZCAPOTZALCO	-1088.054	171.2357	-6.35	0.000	-1423.733	-752.374
BENITO JUAREZ	2353.167	171.2357	13.74	0.000	2017.487	2688.846
COYOACAN	147.0312	171.2357	0.86	0.391	-188.6483	482.7108
CUAJIMALPA DE MORELOS	3204.027	171.2357	18.71	0.000	2868.347	3539.707
CUAUHTEMOC	2254.201	171.2357	13.16	0.000	1918.521	2589.881
GUSTAVO A. MADERO	-1295.324	171.2357	-7.56	0.000	-1631.003	-959.6444
IZTACALCO	-1067.664	171.2357	-6.24	0.000	-1403.343	-731.9842
IZTAPALAPA	-1932.571	171.2357	-11.29	0.000	-2268.251	-1596.892
LA MAGDALENA CONTRERAS	-977.7893	171.2357	-5.71	0.000	-1313.469	-642.1097
MIGUEL HIDALGO	4736.854	171.2357	27.66	0.000	4401.175	5072.534
MILPA ALTA	-3043.482	171.2357	-17.77	0.000	-3379.162	-2707.803
TLAHUAC	-2456.179	171.2357	-14.34	0.000	-2791.858	-2120.499
TLALPAN	-1066.572	171.2357	-6.23	0.000	-1402.251	-730.8924
VENUSTIANO CARRANZA	-547.7626	171.2357	-3.20	0.001	-883.4421	-212.083
XOCHIMILCO	-1868.471	171.2357	-10.91	0.000	-2204.151	-1532.792
_cons	3611.867	121.082	29.83	0.000	3374.505	3849.228

Fig. 8

It is observed in the above regression table that our model remains globally significant, even though our P-value is insignificant in Coyoacan (0.391), which means there isn't enough evidence to conclude that specific delegation has a non-zero effect on the dependent variable. Now that we have defined which are the expensive and not expensive delegations or Alcaldias variables, it is fundamental how other variables seen as externalities influence the price per square meter in gentrified zones and non-gentrified zones, binary dummies were created, as is shown in the following table.

Alcaldía/ delegation	Code (categorical dummies)	Ground price average (per meter square)	Binary dummies (1,0)
Alvaro Obregon	1	3,611 MXN	1
Azcapotzalco	2	2,523 MXN	0
Benito Juarez	3	5,965 MXN	1
Coyoacan	4	3,758 MXN	1

Cuajimalpa de Morelos	5	6,815 MXN	1
Cuahtemoc	6	5,866 MXN	1
Gustavo A. Madero	7	2,316 MXN	0
Iztacalco	8	2,544 MXN	0
Iztapalapa	9	1,679 MXN	0
La magdalena Contreras	10	2,634 MXN	0
Miguel Hidalgo	11	8,348 MXN	1
Milpa Alta	12	568 MXN	0
Tlahuac	13	1,155 MXN	0
Tlalpan	14	2,545 MXN	0
Venustiano Carranza	15	3,064 MXN	0
Xochimilco	16	1,743 MXN	0

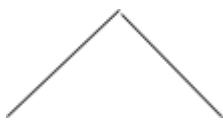
Fig.9

Multivariate Regression

In the previous simple linear regression, we seen that distance from Mexico city zocalo center as the main factor which explains 11.3% of our dependent variable and Alcaldia variable explains 43% of our dependent variable “Y” which mean has a big impact, others variables can be beneficial to analysis this Gentrification phenomenon.

The multivariate linear regression equation:

$$Y = constant + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_7 X_7 + \varepsilon$$

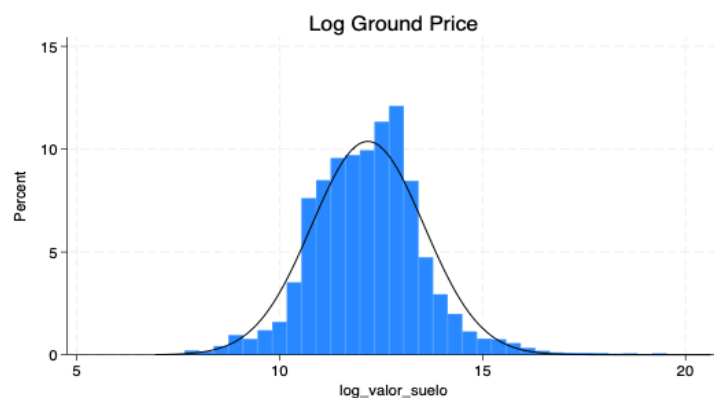
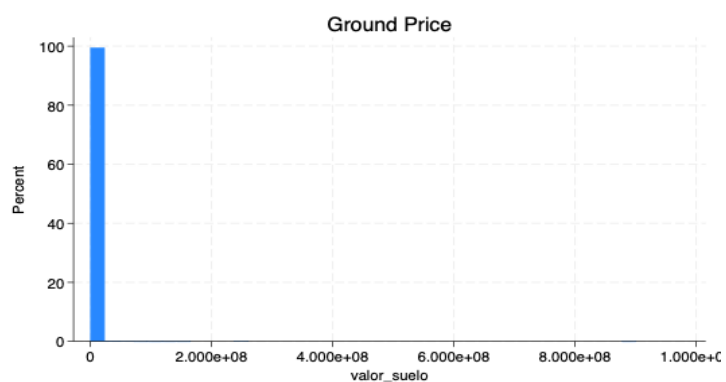
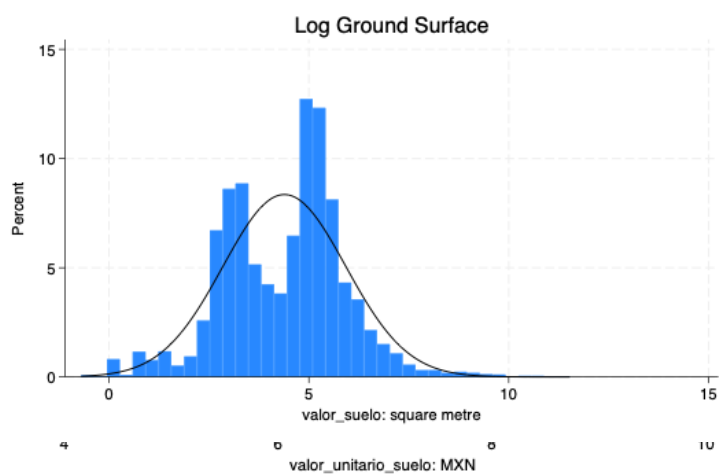
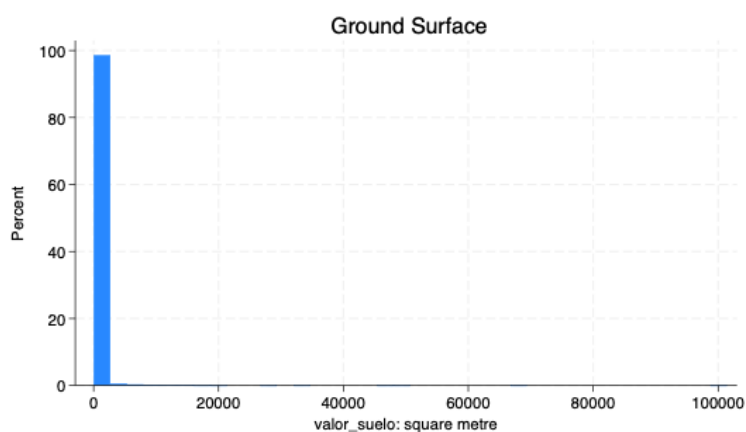
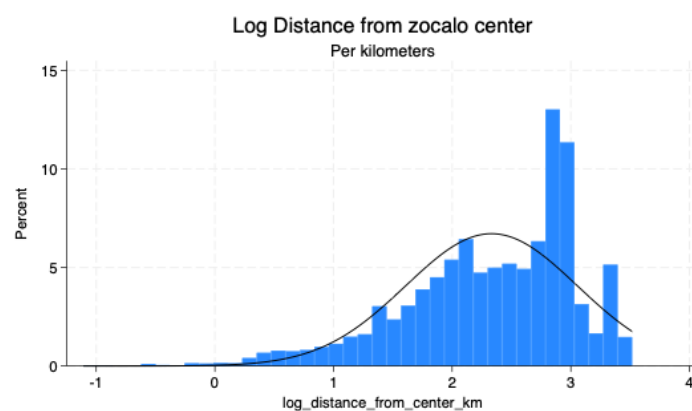
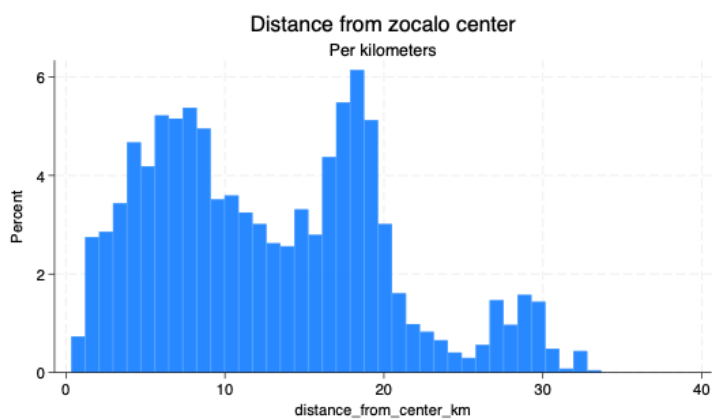


$$\beta_7 X_7 = 0 \quad \beta_7 X_7 = 1$$

Since our regression model is used to measurement of influence of the independent variables (X_i) in the dependent variable (Y), our regression equation is represented above. Each β_i in both regressions is the coefficient of that corresponding X_i and ε is the residual or the error term. In the variable alcaldia binary dummies were created, $X_i = 1$ represents the slope for expensive delegations for locals and $X_i = 0$ represent the no expensive delegations for mexican locals.

I. Level data vs. log data

Below are the variables used in the regression and their respective distribution shape:



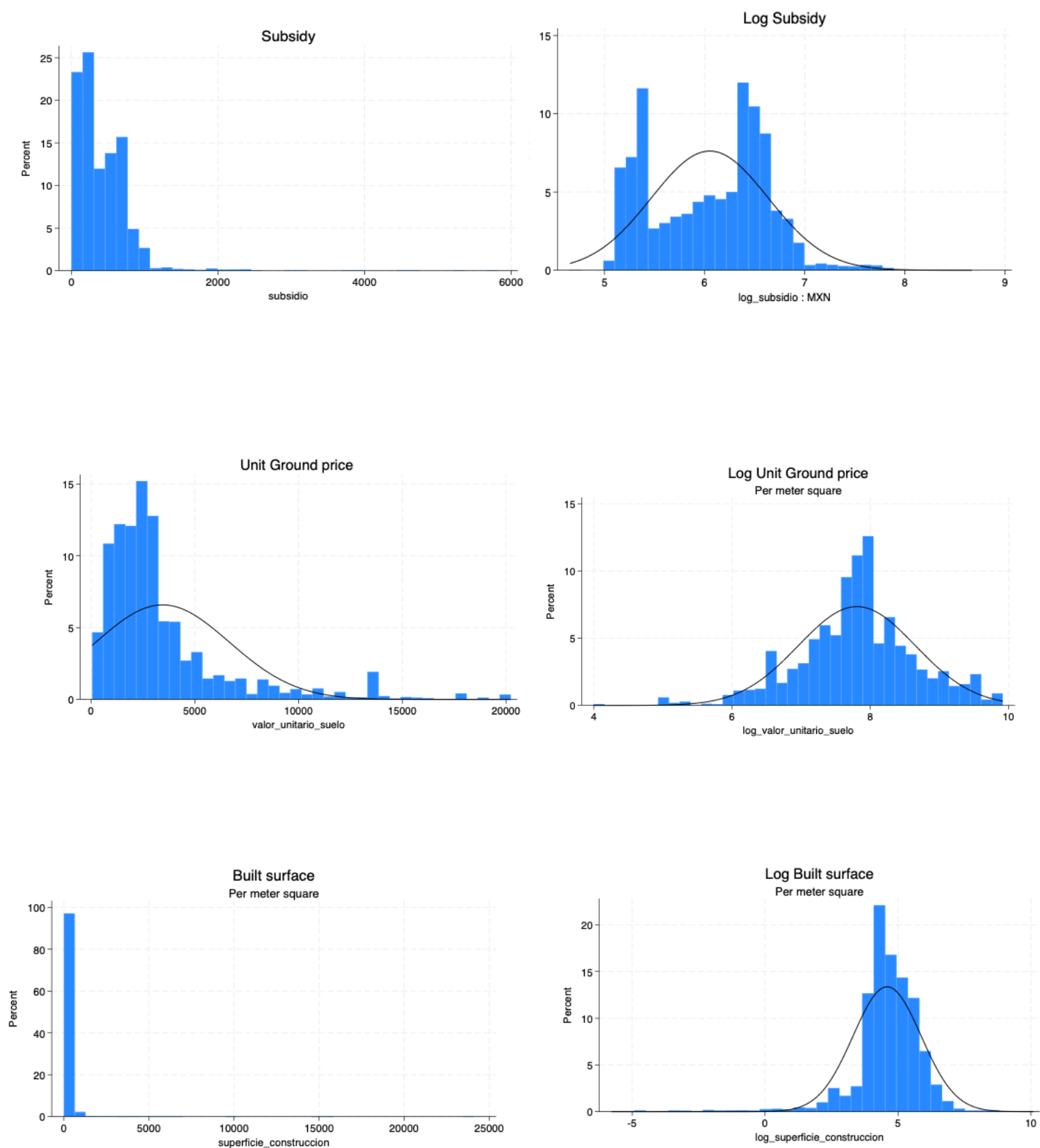


Fig. 10

It was necessary to keep the previous remaining variables after the cleaning data process, so their respective parameters can be interpreted as percentages or elasticities of our interest in further multivariable analysis. The logged data is that it will make our residual look “normal” distribution, reduce heteroskedasticity and diminish the probability to find outliers with significant values during our regression analysis. In all the level data diagrams (with expectation of year built) you can see that the level data are mostly skewed left and logging these data makes them more symmetrically distributed and improves the model’s fit you can notice that the longer bars that come out of the normal distribution are compensated by the empty spaces of the normal distribution. However, it is still missing to run both models to identify their R-squared, in this way we can choose the most appropriate, as well as we will have to leave year built ([anio_construccion](#)) variable as level data.

Source	SS	df	MS	Number of obs	=	6,400
Model	1.0185e+10	6	1.6975e+09	F(6, 6393)	=	193.85
Residual	5.5982e+10	6,393	8756787.72	Prob > F	=	0.0000
Total	6.6167e+10	6,399	10340197.6	R-squared	=	0.1539
				Adj R-squared	=	0.1531
				Root MSE	=	2959.2

valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
valor_suelo	.0000276	4.55e-06	6.08	0.000	.0000187	.0000365
distance_from_center_km	-151.0867	5.099403	-29.63	0.000	-161.0833	-141.0902
subsidio	-.9324739	.0936066	-9.96	0.000	-1.115974	-.7489736
anio_construccion	25.98086	2.550389	10.19	0.000	20.98124	30.98048
superficie_construccion	.2795795	.0920906	3.04	0.002	.0990509	.460108
superficie_terreno	-.1406998	.0230965	-6.09	0.000	-.1859768	-.0954229
_cons	-45989.98	5070.586	-9.07	0.000	-55930.03	-36049.94

Fig. 11. the variable unit ground price ([valor_unitario_suelo](#)) being the dependent variable.

Source	SS	df	MS	Number of obs	=	6,400
Model	9.8616e+09	5	1.9723e+09	F(5, 6394)	=	223.97
Residual	5.6305e+10	6,394	8805968.95	Prob > F	=	0.0000
Total	6.6167e+10	6,399	10340197.6	R-squared	=	0.1490
				Adj R-squared	=	0.1484
				Root MSE	=	2967.5

valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
distance_from_center_km	-153.0763	5.103147	-30.00	0.000	-163.0802	-143.0724
subsidio	-.9396197	.0938617	-10.01	0.000	-1.12362	-.7556194
anio_construccion	26.71161	2.554695	10.46	0.000	21.70355	31.71967
superficie_construccion	.5625367	.0796681	7.06	0.000	.4063605	.7187128
superficie_terreno	-.0395729	.016057	-2.46	0.014	-.07105	-.0080958
_cons	-47469.76	5078.935	-9.35	0.000	-57426.17	-37513.34

Fig. 12. removing the variable ground price ([valor_suelo](#)), just leaving unit ground price ([valor_unitario_suelo](#)) as dependent variable.

Source	SS	df	MS	Number of obs	=	4,916
Model	2714.7734	6	452.462233	F(6, 4909)	>	99999.00
Residual	.008886643	4,909	1.8103e-06	Prob > F	=	0.0000
				R-squared	=	1.0000
				Adj R-squared	=	1.0000
Total	2714.78229	4,915	.552346345	Root MSE	=	.00135

log_valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
log_valor_suelo	.999956	.0000425	2.4e+04	0.000	.9998727	1.000039
log_distance_from_center_km	-.0000802	.0000357	-2.24	0.025	-.0001503	-.0000102
log_subsidio	-.0004558	.0000562	-8.11	0.000	-.0005659	-.0003456
anio_construccion	-1.30e-06	1.46e-06	-0.89	0.375	-4.17e-06	1.57e-06
log_superficie_construccion	.0005581	.0000414	13.47	0.000	.0004769	.0006393
log_superficie_terreno	-.9999952	.0000363	-2.8e+04	0.000	-1.000066	-.999924
_cons	.0034065	.0029652	1.15	0.251	-.0024067	.0092197

Fig. 13. the variable log ground unit price ([log_valor_unitario_suelo](#)) being the dependent variable and keeping variable ground price ([valor_suelo](#)).

Source	SS	df	MS	Number of obs	=	4,916
Model	1711.37006	5	342.274012	F(5, 4910)	=	1674.85
Residual	1003.41222	4,910	.204360942	Prob > F	=	0.0000
				R-squared	=	0.6304
				Adj R-squared	=	0.6300
Total	2714.78229	4,915	.552346345	Root MSE	=	.45206

log_valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
log_distance_from_center_km	-.318716	.0111122	-28.68	0.000	-.3405009	-.2969311
log_subsidio	.7415664	.0156254	47.46	0.000	.7109337	.772199
anio_construccion	-.0069977	.000482	-14.52	0.000	-.0079426	-.0060528
log_superficie_construccion	-.1988882	.0136236	-14.60	0.000	-.2255965	-.1721799
log_superficie_terreno	-.3645922	.0081763	-44.59	0.000	-.3806215	-.3485629
_cons	20.35441	.9530146	21.36	0.000	18.48608	22.22275

Fig. 14. the variable log ground unit price ([log_valor_unitario_suelo](#)) being the dependent variable and deleting variable ground price ([valor_suelo](#)).

The first regression output of unit ground price ([valor_unitario_suelo](#)) representing the dependent variable, exactly as we did previously in the simple regression model gave a R-squared of 15.3%, which means that independent variables only explains 15.3% of the dependent variable, while for the second regression (see table 2) the variable ground price ([valor_suelo](#)) had to be removed due to avoid correlation issues later, and the R-squared gets down 14.9%. Since previously we had already verified that log data have a better shape distribution, another regression model was run with log data, including all variables and gave us a R-squared of 1.000, which is a completely error, this result could have been caused because the price per square meter is included in total price variable ([valor_suelo](#)), in the fourth run regression output we got 63% with log data and total price variable removed. Comparing the fourth regression results from the previous three tables, it's evident that log data regression (table 4) performs better than the level data using the variable unit ground price ([valor_unitario_suelo](#)) in which logged data variables superpasses with 63% of R

square on explanation, for this reason our last model is the one we will continue using for the analysis.

In the fourth previous tables, the regression was conducted on all the independent variables and it can be seen that Prob> F= 0.000 indicating that the regression model is globally significant meaning that at least one of the independent variables in our model has a significant effect on the dependent variable, in the same way you can notice that in level data (the first two tables) the variables distance from center ([distance_from_center_km](#)) and subsidies ([subsidio](#)) has a negative correlation in relation with their respective dependent variable, since the more the distance of the CDMX zocalo the lower is value the ground per total square or total price, while higher government subsidies are given to lower house pricing. In all regressions, the variable ground surface ([superficie_terreno](#)) has a negative correlation with the dependent variable in level data and log data, because the more the ground surface increases, the price per meter square decrease, comparing the households located in surrounding delegations than small departments located in CDMX downtown.

The negative correlation of variable log year build ([log_anio_construccion](#)) is due to the data is not being logged, since it is better if we take it as string rather than numerical, so we don't have to include it in our log data model, even our local P-value says is less than 0.05.

II. Checking for correlation

Previously to continue with the analysis is substantial to check the correlation between our Independent variables to avoid a high correlation which leads us to having multicollinearity issues. In that case if there is high correlation it can be difficult to separate the independent variables effects on the dependent variable provoking that the coefficients of the correlated variables may be imprecisely estimated. From the table below you can see four possible multicollinearity issues:

The first one is a moderate negative correlation between logarithm of distance ([log_distance_from_center_km](#)) and unit ground price ([log_valor_unitario_suelo](#)) with a correlation coefficient of -0.5935. Then we have a moderate negative correlation between unit ground price ([log_valor_unitario_suelo](#)) and ground surface ([log_superficie_terreno](#)) with a correlation coefficient of -0.5383.

There is a moderate positive correlation between distance from center ([log_distance_from_center_km](#)) and ground surface ([log_superficie_terreno](#)) with a correlation coefficient of 0.4954, as total surface area increase, the distance from center tends to increase and finally it was found a moderate positive correlation between ground surface ([log_superficie_terreno](#)) and built surface ([log_superficie_construccion](#)) with correlation coefficient of 0.5260, as built surface increase, the ground surface area also tends to increase. You can see that the rest of the correlations are weaker and closer to zero, indicating little to no linear relationship between those variables.

	log_valor_unitario_suelo	log_distance_from_center_km	log_superficie_terreno	log_anio_construccion	log_superficie_construccion	log_superficie_terreno
log_valor_unitario_suelo	1.0000					
log_distance_from_center_km	-0.5935	1.0000				
log_superficie_terreno	0.2178	0.0376	1.0000			
log_anio_construccion	-0.0346	0.1015	0.0105	1.0000		
log_superficie_construccion	-0.0905	0.1236	0.7065	-0.0920	1.0000	
log_superficie_terreno	-0.5383	0.4954	0.4091	-0.1907	0.5260	1.0000

Fig. 14

III. Checking Multicollinearity

To verify those four previous correlations should be taken into account when analyzing the data, as multicollinearity can affect the results of regression analysis, is better to test multicollinearity; The variance Inflation Factor (VIF) is computed to test multicollinearity of the first estimation model (the level data), in case there exist multicollinearity, adjustment will be made to correct it, since we have to corrected it to avoid problems with the interpretation of the coefficients. In the case that VIF value is larger than 10, it is necessary to consider removing the predictor, as you can see in the table below, that all variables have VIF values less than 5 indicating there is no multicollinearity existing in the model. Hence any adjustment will be made under this step.

Variable	VIF	1/VIF
log_superf~n	2.32	0.430124
log_subsidio	2.08	0.480305
log_superf~o	2.06	0.486147
log_distan~m	1.49	0.671965
anio_const~n	1.13	0.888615
Mean VIF	1.82	

Fig. 15

IV. Testing Heteroskedasticity

Heteroskedasticity was checked in two different ways, using the “rvfplot” graph, where residuals are plotted against the fitted values of dependent variables and examine the pattern of residuals plotted against the predicted values of the dependent variable. Any pattern of the residuals indicates the existence of heteroskedasticity but if there is no patten and the plot is uniformly distributed, then it means there is no heteroskedasticity.

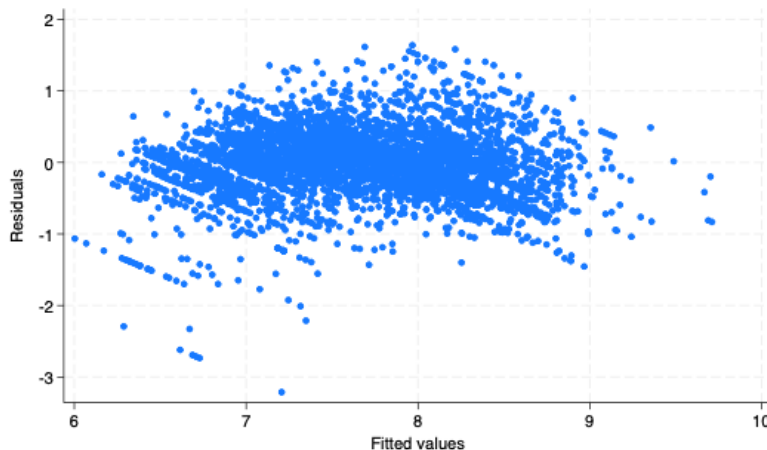


Fig. 16

In the graph above, we can observe the trend in residuals and the spread of residuals, in this plot there seems to be a slight downward trend in the residuals as the fitted values increase. This can indicate a model misspecification where the relationship between the dependent and independent variables is not entirely linear or an important variable is omitted. While the spread of residuals does not appear to be constant across the range of gifted values, specifically at lower values (around 6 and 7) the residuals show a wider spread, and as the

fitted values increase (from 7 to 10), the spread of the residuals appears to become narrower.

```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: i.i.d. error terms
Variables: All independent variables

H0: Constant variance

F(5, 4910) = 24.55
Prob > F = 0.0000

```

Fig.17

In this second method, we had to check what was already evident. In the above Breusch-Pagan/ Cook-Weisberg test for heteroskedasticity, it shows that the test statistic follows an F-distribution with 5 and 4910 degrees of freedom(for the numerator and denominator). The F- value in this case is 24.55 which suggests a significant deviation from homoscedasticity or means that the variation in the error term is significantly related to the independent variables in the model, while the P-value (Prob > F) which indicates the probability of observing an F-statistic the one observed under the assumption of null hypothesis, in this case the p-value is effectively zero (0.0000) indicating that the probability of obtaining such an F value under the null hypothesis is extremely low, or in other words the p-value is lower than any conventional significance level (e.g 0.05, 0.01 or 0.001) and thus the null hypothesis of homoscedasticity is rejected.

V. Correcting Heteroskedasticity

The previous pattern in the graph suggests the presence of heteroskedasticity, where the variance of the residuals is not constant across all levels of fitted values. This non - constant variance violates the assumption of homoscedasticity, which states that the variance of the residuals should be across all levels of the independent variables. To correct this heteroskedasticity issue, I made use of Robust standard errors, which adjust the standard errors of the regression coefficient without changing the coefficients themselves.

```

Linear regression               Number of obs   =    4,916
                               F(5, 4910)         =   1300.07
                               Prob > F           =    0.0000
                               R-squared           =    0.6304
                               Root MSE        =    .45206

```

log_valor_unitario_suelo	Robust				
	Coefficient	std. err.	t	P> t	[95% conf. interval]
log_distance_from_center_km	-.318716	.0105457	-30.22	0.000	-.3393902 -.2980417
log_subsidio	.7415664	.0177261	41.83	0.000	.7068153 .7763174
anio_construccion	-.0069977	.0004818	-14.52	0.000	-.0079422 -.0060532
log_superficie_construccion	-.1988882	.014587	-13.63	0.000	-.2274852 -.1702913
log_superficie_terreno	-.3645922	.0084391	-43.20	0.000	-.3811367 -.3480477
_cons	20.35441	.9608065	21.18	0.000	18.4708 22.23803

Fig. 18 Robust Regression Result

	robust b/se	ols b/se
log_distan~m	-.318716*** .0105457	-.318716*** .0111122
log_subsidio	.7415664*** .0177261	.7415664*** .0156254
anio_const~n	-.0069977*** .0004818	-.0069977*** .000482
log_superf~n	-.1988882*** .014587	-.1988882*** .0136236
log_superf~o	-.3645922*** .0084391	-.3645922*** .0081763
_cons	20.35441*** .9608065	20.35441*** .9530146

Fig. 19 Robust Regression vs OLS

The above table presents the results of a regression analysis, comparing two models: one with robust standard error and the other with ordinary least squares (OLS) standard errors. We can see that the coefficient values of the OLS model are the same as the coefficient of the robust, but the values of the standard error which are below each coefficient are different. In robust regression, the standard errors are adjusted to account for potential heteroskedasticity and the presence of outliers, providing more reliable estimates of the variability in the parameter estimates.

The results of the table can be interpreted as in the first variable logarithm of distance ([log_distance_from_center_km](#)) a 1% increase in distance is associated with approximately a 0.319% decrease in the dependent variable, holding other factors constant. The variable logarithm of subsidy ([log_subsidio](#)) is significant at 1% ($p < 0.01$), as 1% increase in subsidy, approximately 0.742% increase in the dependent variable, holding other factors constant. In the variable year of construction ([log_anio_construccion](#)), each additional year is associated with approximately a 0.007% decrease in the dependent variable, while the variable logarithm built surface ([log_superficie_construccion](#)) a 1% increase of build surface area is associated with approximately a 0.199% decrease in the dependent price variable. Finally, the variable logarithm ground surface ([log_superficie_terreno](#)) a 1% of ground surface area is associated with approximately 0.365% in the dependent variable. The constant is the intercept of the regression line. It represents the expected value of the dependent variable when all independent variables are zero, but in this case since it is a logarithmically model, this means variables are equal to one.

VI. Selecting variables

To be completely sure about selecting the most important variables in order to get the best model for further analysis, stepwise regression was used, which involves the selection of independent variables to be used in the final model. The rejection would start including all possible variable predictors and give the result that all variables predictors included in the model indicates that p-values from the wald test are less than 0.05 so this means there is strong evidence to reject the null hypothesis that the coefficient of that predictor is zero and there not immediate need to remove any predictor from the model due to insignificance, ensuring that all included predictors are meaningful and should be part of the final model.

Wald test, begin with full model:
 $p < 0.0500$ for all terms in model

Source	SS	df	MS	Number of obs	=	4,916
Model	1711.37006	5	342.274012	F(5, 4910)	=	1674.85
Residual	1003.41222	4,910	.204360942	Prob > F	=	0.0000
				R-squared	=	0.6304
				Adj R-squared	=	0.6300
Total	2714.78229	4,915	.552346345	Root MSE	=	.45206

log_valor_unitario_suelo	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
log_distance_from_center_km	-.318716	.0111122	-28.68	0.000	-.3405009	-.2969311
log_subsidio	.7415664	.0156254	47.46	0.000	.7109337	.772199
anio_construccion	-.0069977	.000482	-14.52	0.000	-.0079426	-.0060528
log_superficie_construccion	-.1988882	.0136236	-14.60	0.000	-.2255965	-.1721799
log_superficie_terreno	-.3645922	.0081763	-44.59	0.000	-.3806215	-.3485629
_cons	20.35441	.9530146	21.36	0.000	18.48608	22.22275

Fig. 20

VII. Checking endogeneity

There might be endogeneity issues when the independent variables are correlated with the error term of the regression model. Logically, there tend to be higher built area surface from larger ground area surface, but considering that many of Mexico city downtown buildings are located in a narrow ground area surface compared with bigger houselivings areas on the outskirts of the city. The 2 stage least square (2SLS) regression to assess the degree of endogeneity and identify which variables are endogenous.

The Durbin-Wu-Hausman test evaluates the null hypothesis that the regressors are exogenous. In other words, if the p-value is low, it suggests that the null hypothesis can be rejected, indicating the presence of endogeneity. The chi-square is 0 with a p-value that is not provided indicated as “.”, this would mean that the test statistic is too small to reject the null hypothesis of exogeneity. And the wu- Hausman F-statistic and p-value indicated as “.” are not provided as well, suggesting that the test doesn't detect endogeneity in the regressors. In this case it is not necessary to modify anything since in 2SLS regression and Durbin-Wu-Hausman test indicated no endogeneity issues.

Instrumental variables 2SLS regression

Number of obs = 4,916
Wald chi2(5) = 8384.49
Prob > chi2 = 0.0000
R-squared = 0.6304
Root MSE = .45179

log_valor_unitario_suelo	Coefficient	Std. err.	z	P> z	[95% conf. interval]	
log_distance_from_center_km	-.318716	.0111054	-28.70	0.000	-.3404822	-.2969498
log_subsidio	.7415664	.0156158	47.49	0.000	.7109599	.7721728
anio_construccion	-.0069977	.0004817	-14.53	0.000	-.0079418	-.0060536
log_superficie_construccion	-.1988882	.0136153	-14.61	0.000	-.2255736	-.1722028
log_superficie_terreno	-.3645922	.0081714	-44.62	0.000	-.3806078	-.3485766
_cons	20.35441	.9524329	21.37	0.000	18.48768	22.22115

(no endogenous regressors)

. estat endog

Tests of endogeneity

H0: Variables are exogenous

Durbin (score) chi2(0) = 0 (p = .)

Wu-Hausman F(0,4910) = . (p = .)

Fig. 21

VIII. Checking linearity

Linearity is important because if the relationship is not linear, the linear regression model will not fit the data well, and the estimates of the coefficients and the predictors will be biased and inaccurate. You can see in the below graphs that only the variable distance from center has “U” shape, the majority of the dots are located in left side of the graph, leaving various outliers in the opposite side, I tried to fix this issue dropping those outliers but the “U” shape remains the same, so it was necessary to apply logarithmic transformation to either or both the dependent and independent variables, it was the only way to fulfill with the linearity property.

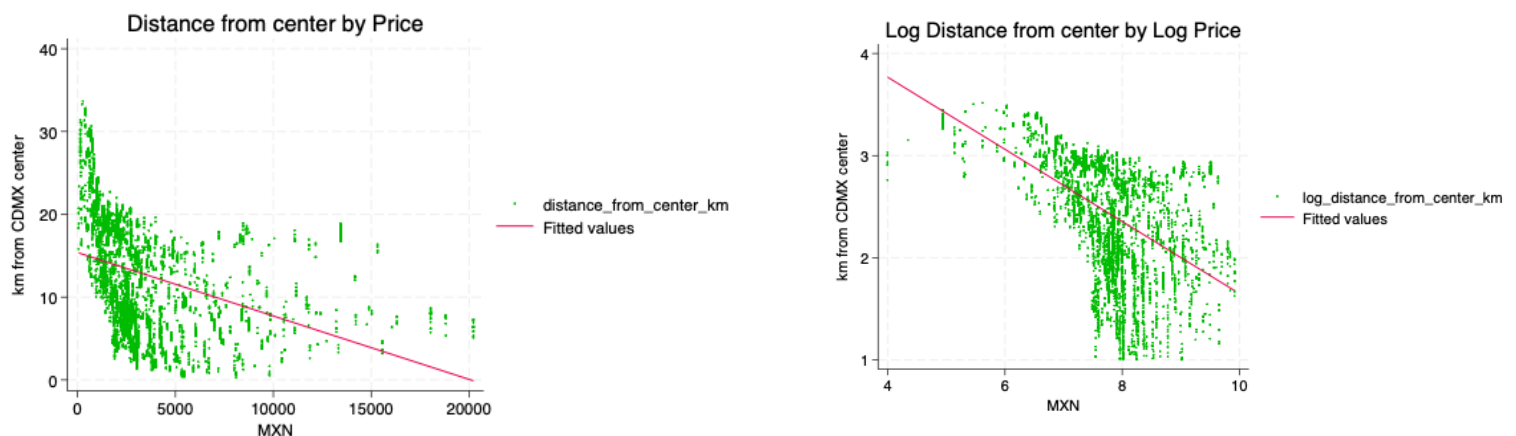


Fig. 22

Today, Mexico city is consider by various studies to be the be worst city in the world to move around because the travel transportation times are very high. Mexico city residents in search of accessible housing must invest up to four hours in public or private transportation and



around 30% and 45% of their economic income in transportation tickets or gasoline according with the Mexican Institute of Competitiveness (IMCO)⁸. Otherwise, more than half of households whether houses or departments were concentrated in outskirts. The rise in real estate prices in Mexico city downtown has led residents to see a viable solution not only the outskirts of Mexico city (pink color on the map), but also in other closer states, specifically, the State of Mexico (orange on the map) and Toluca (green on the map) had become an important development point for real state projects. The real estate valuation and consulting firm Tinsa Mexico reports that of the 6,500 homes placed in 2023, 44% correspond to Mexico City and 56% to the suburban

municipalities of State of Mexico and Hidalgo⁹. The state of Mexico is located around 1 to 2 hours from Mexico city center, while Hidalgo is around 2 to 3 hours by private transportation, being distance an important factor in the real state market.

In the second set of twoway scatter graph you can see that our fitted values line has slightly zero slope in the left graph, hence we can say that the built surface area isn't impacted by the unit value per meter square, although there so many outliers that represents the maximum built surfaces, trimming those outliers and transforming data to logarithmic then we get data clustered in smaller lower bandwidth than the previous one without any transformation. In the right graph you can see a slightly negative correlation between x-axis and y-axis, meaning that more the constructed area increases, a lower price per square meter, although this seems illogical, this phenomenon is due to the location added value.

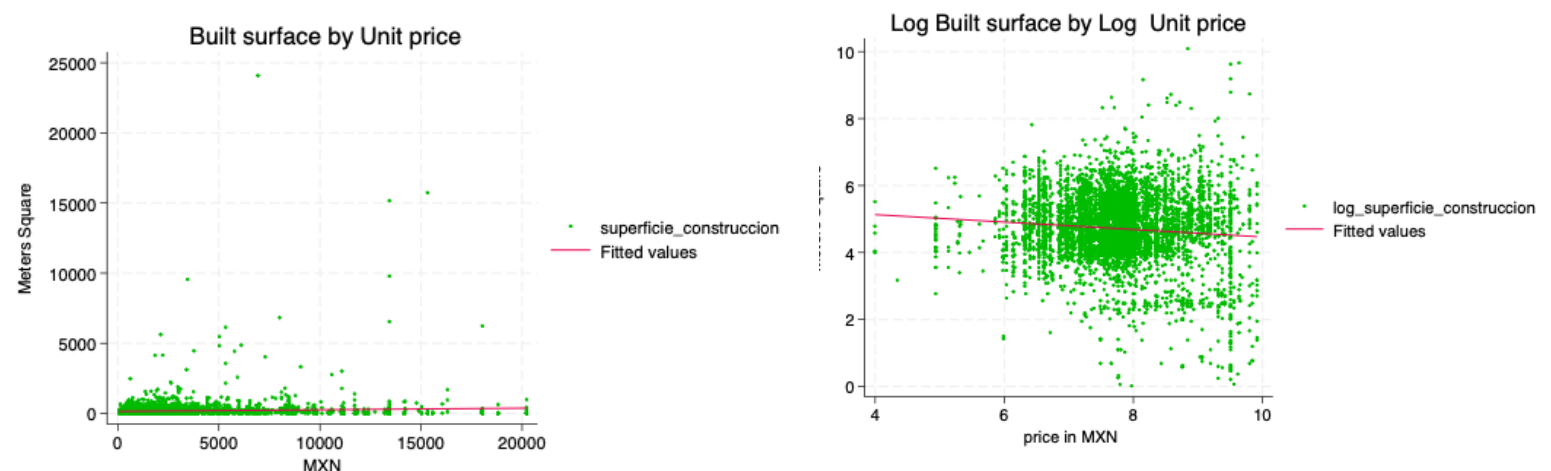


Fig. 23

⁸ According with Excelsior "The costs of living in the periphery, they lose 4 hours and 45% of income" <https://www.excelsior.com.mx/comunidad/2016/11/21/1129441> (2016) accessed June 20th 2024.

⁹ According with Infobae Mexico "Demand for housing on the outskirts of CDMX grows due to increased costs in central areas" <https://www.infobae.com/mexico/2024/02/18/crece-demanda-de-viviendas-en-periferia-de-la-cdmx-de-bido-al-incremento-de-costos-en-zonas-centrales/> (2024) accessed June 20th 2024.

For instance, not long ago Mexico City faced a very serious aesthetic problem, due to lack of urban planning, in which was reflected the bias between different social classes, the rural area of the periphery was replaced by houselivings residents without any green area for the working class, sharpening the distinction among residence such as “Lomas de Chapultepec” or “Jardines del pedregal” for elite class and “Casas GEO” for working class. In recent years a project to combat this problem was the construction of “Satélite” cities built on a territory that was previously a garbage dump or other insignificace place where large corporate buildings and luxury apartments were built without considering they were around self-construction houselivings for lower class, this resulted in the coexisting of irregular self-construction settlements where living conditions are very different and whose aesthetics have nothing related to buildings that rise dozens of meters above, triggering the construction of fence due to insecurity reasons¹⁰.

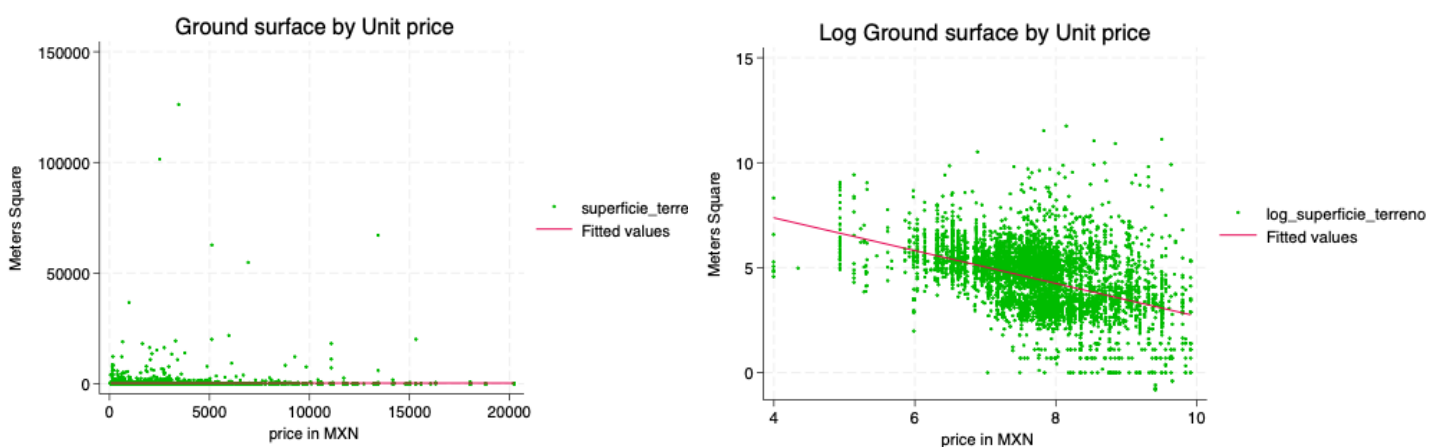


Fig. 24

In the same way, the correlation between the ground surface by unite price has cero slope and the dots are cluster below left of the graph but you can see a perfect correlation; while in the logarithmic graph you can see a significant negative correlation and observations are clustered in a compact bandwith, the larger the ground area lower the price per square meter is. This is due to what had previously been mentioned, the construction area or the ground surface area is influenced by the location added value. A clear example of this case is Milpa Alta two hours far away from Mexico center in private transportation, which is observed in the single linear regression Milpa Alta is one of the chepeast delagation to get a pice of land, but this delegation has not added value, since is a place committed to defending and preserving the mother indigenous tongue “Nahuatl” and its main economy unlike the rest of Mexico city is based on agricultural sector the production of vegetables, nopal, etc. That is why, this negative correlation, buying land in this delegation, no matter

¹⁰ According with Arch daily “The aestheticization of inequality: contrasting landscapes on the outskirts of Mexico city” <https://www.archdaily.mx/mx/983054/la-estetizacion-de-la-desigualdad-paisajes-de-contraste-en-la-periferia-de-la-ciudad-de-mexico> (2022) accessed June 20th 2024.

how large it is, is far from the industrialized, modern and corporate side for which Mexico city is known¹¹.

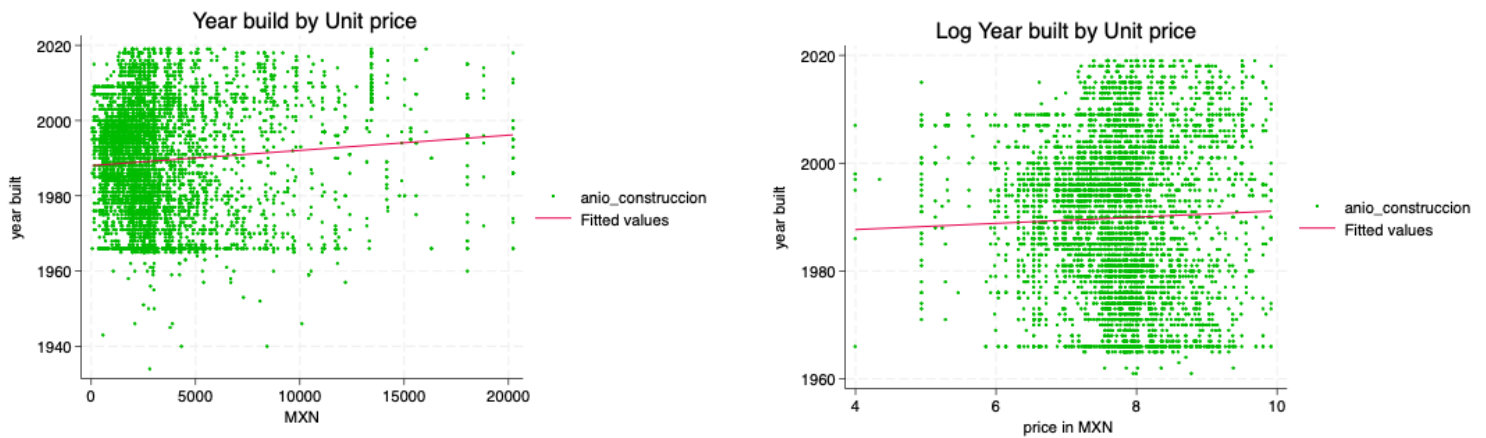


Fig. 25

In the fourth set of twoway scatter graphs between the year built by price per meter square is seen a positive correlation, in both graphs, with level data and log data, since the fitted value line has a slightly positive slope, is evident that higher prices per meter square are, most recent are the houselivings construction are. In both graphs is observed that observations are clusters in a bandwidth between 1970 and 2020, with a mean of 1988, which makes sense since many of the them were reconstructed or fixed after the earthquake in 1985 which reached a magnitud of 8.1 degrees in Richter scale¹². In this case it doesn't matter how modern is the property due to it most to fulfill with building codes and safety protocols to avoid catastrophic damage when an earthquake occurs. You can notice as well that the outliers on right up side have a hger price per meter square in contrast with the outliers in the left down side, which are houselivings builted previously 1960 and have a cheaper price per meter square. For the right graph with log data it was necessary to drop observations smaller before 1960.

¹¹ According with Noticias digitales y periodistas desde el Zocalo de la Ciudad de Mexico "Vanegas Tapia commits to preserving the Nahuatl language in Milpa Alta"
<https://revistazocalo.com/se-compromete-vanegas-tapia-a-preservar-la-lengua-nahuatl-en-milpa-alta/> (2021) accessed June 24th 2024.

¹² According with Government of Mexico "Earthquake, Mexico 1985"
<https://www.gob.mx/siap/articulos/terremoto-mexico-1985?idiom=es> (2020) accessed June 24th 2024.

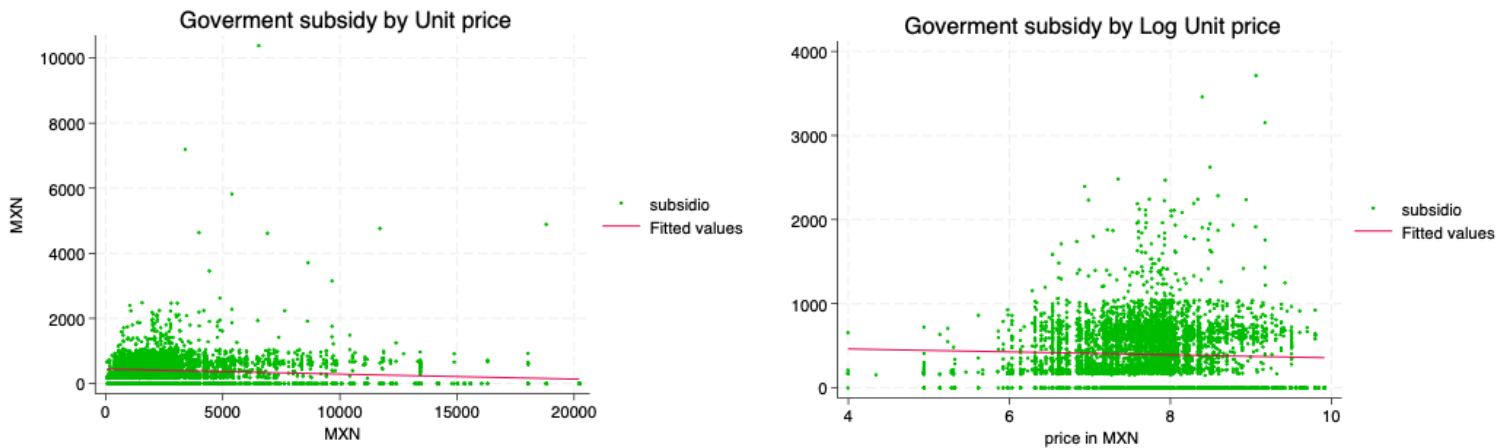


Fig. 26

Finally in the fifth set of twoway scatter graphs you can see that in the left graph, the observations are clustered on above left side of the graph, showing that the government subsidies are clustered in favor of those houselivings which the price per meter square is less than 5000 MXN (an equivalent of 250 EUR), meanwhile the mean of these subsidies are 388 MXN (an equivalent of 20 EUR). There is an outlier on the top of the left graph which represent the maximum subsidy of 10,379 MXN (around 524 EUR) and few outliers on the right below side representing that houselivings valued around 15,000 MXN (around 700 EUR) per meter square, the inclination of the correlation is slightly negative, which make sense since the subsidies are granted for those houselivings value at lower price per meter square. While in the logarithmic transformation graph, it was necessity to trim outliers greater than 4000 MXN in subsidy variable, in order to get a compact bandwidth in which the mean observations are clustered. One of these subsidies is the social program houseliving improvement or “Meoramiento de vivienda”, with these resources, which imply an investment of more than 243.4 million MXN, 1,822 families are benefited with resoruces to carry out maintenances or improvement work in their home, as well as build new homes on family land lots. In addition, with the subsidies granted to them, they will be able to install eco-tecnological devices such as solar heaters, which result in financial savings when paying for services such as gas¹³.

IX. Final Model

The multivariate linear regression equation:

$$Y = constant + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_7 X_7 + \varepsilon$$

$$\begin{array}{ccc} & \wedge & \\ \beta_7 X_7 = 0 & & \beta_7 X_7 = 1 \end{array}$$

¹³ According with Mexico City Housing Institute “Derivery of 1,822 loans and subsidies from the Housing Improvement program”
<https://www.invi.cdmx.gob.mx/comunicacion/nota/EntregaDe1822CreditosYSubsidiosDelProgramaDeMejoramientoDeVivienda> (2023) accessed June 24th 2024.

Since our regression model is used to measurement of influence of the independent variables (X_i) in the dependent variable (Y), our regression equation is represented above. Each β_i in both regressions is the coefficient of that corresponding X_i and ϵ is the residual or the error term. In the variable *alcaldia* binary dummies were created, $X_i = 1$ represents the slope for expensive delegations for locals and $X_i = 0$ represent the no expensive delegations for mexican locals.

With all the adjustment above, the final model would be the following, in which shows an R-square of 66.02%, 3% better than the initial multivariate regression (see figure 14) with an R-square of 63%, the final model was run twice, when the dummy variable (*affordable_alcaldia*) is 1 and 0, a noticeable change was observed when (*affordable_alcaldia*) is 0, since the R-square get 61.9% taking into account in the regression the non expensive delegations or *alcaldas*. The model is global significant.

```
. reg log_valor_unitario_suelo log_distance_from_center_km log_subsidio anio_construccion log_superficie_construccion log_superficie_terreno
> i.affordable_alcaldia, robust
```

```
Linear regression      Number of obs   =    4,630
                      F(6, 4623)       =   1182.08
                      Prob > F         =    0.0000
                      R-squared        =    0.6602
                      Root MSE      =    .43154
```

		Robust				
log_valor_unitario_suelo	Coefficient	std. err.	t	P> t	[95% conf. interval]	
log_distance_from_center_km	-.3552423	.0124308	-28.58	0.000	-.3796125	-.3308872
log_subsidio	.6386554	.0190673	33.49	0.000	.6012743	.6760364
anio_construccion	-.0072152	.0004714	-15.31	0.000	-.0081394	-.0062911
log_superficie_construccion	-.1848494	.0159404	-11.60	0.000	-.2161002	-.1535987
log_superficie_terreno	-.3097869	.0086841	-35.67	0.000	-.3268119	-.292762
1.affordable_alcaldia	.3564916	.0147568	24.16	0.000	.3275612	.3854219
_cons	21.09362	.93807	22.49	0.000	19.25456	22.93269

Fig. 27

Conclusion

It is evident that gentrification is playing a crucial role in the valorization of certain areas of the city. In municipalities like Cuahatemoc, the phenomenon of gentrification is driving an increase in prices due to development of real estate projects and foreign investments. However, this process is also generating adverse effects, such as the displacement of local population and loss of cultural identity in historical colonies.

It is also visible how gentrification is related to perceived safety and urban improvements in the delegations mentioned previously. While it is encouraging to see investments in infrastructure and services to reduce crime, we must question who really benefits from these improvements and who is left out. Since our data used for this analysis is a cross-sectional data, recovered from 2022 it would be fundamental to have a time series data, in which we can analysis to completely assure the settlement of foreigners to Mexico City along with the

increase price in houselivings in the real state sector. One of the recent solutions that the Deputy Chamber has approved is the taxation on digital nomads of \$103.4 pesos per night based on Measurement and Update Unit (UMA) in addition to the originally contracted term.

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